Application of Blockchain Technology in Travel Industry

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***Abstract*— A Blockchain is basically a decentralized, distributed ledger of all the transactions or events which takes place within a network that involves multiple parties. The transactions that happen are anonymous hence it has very high level of security. Any transaction that takes place is verified by all the members present in the chain. However, there are some drawbacks as well like scalability. This technology can be used in different areas and one of them is the travel industry which can change the whole transaction dynamics of the traditional way the travel sector works. This paper will give the outline or an idea of the underlining concepts about this new technology and how it can be used in travel industry.**

I. Introduction

Blockchain technology also called as a distributed ledger technology which acts as a data structure is referred to as blocks in a chain, where the value of each block is referred by the block prior to them. After creating the blockchain and the information is fed, it is not possible to tamper the network to get the details which is shared to all the members across the network. All the users in a blockchain are aware of all the transactions that take place. A blockchain contains blocks, wherein every block is the successor of the previous block, it uses the nonce and signature of the previous block as a key to go forward to the next block. A digital signature plays a major role as the transactions are broadcasted across the public network. As the public network is not safe and the details can be easily tampered, the digital signature contains a signing phase and a verification phase to prevent fraud using private key and public key. The private key is a confidential key which is known only to the user to sign the transactions and the public key is known to both the users. A blockchain system are segregated roughly into private and public blockchain. In public blockchain anyone can read and write data into the ledgers. Whereas in private blockchain only the trusted participants can read and write data into the ledgers. Hence it is defined as a secured distributed public/private ledger which can store data over a peer-to-peer network.

Figure 1 represents the basic flow of blockchain wherein it contains a genesis block which is the first block that contains the genesis transaction details and the very first hash value. This block is referred as i-1th Block. The second block i.e the ith Block contains transaction details and the hash value of the previous block i.e the genesis block. The third block is referred as i+1th Block, which is the consecutive block containing the transaction details and the previous hash value. Sequence of blocks are constructed in the similar manner to form a blockchain. A blockchain can contain only one genesis block. The size of the block and size of each transaction is used to determine the maximum number of transaction that a block can contain within itself.

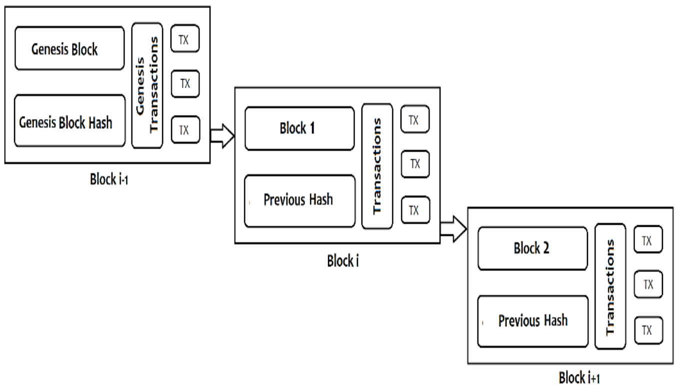

Figure 1: basic flow of blockchain

The main objective is to implement this architecture into the travel industry where all the transaction data is secure and can be verified by all the members in the network which makes it more reliable. Another reason to use blockchain in travel is that the use of cryptocurrency can be more effectively implemented in this sector. It involves airline booking, hotel booking, rail booking, any sort of travel booking which people do these days instead of personally go and buy the ticket. Everything is digitalized and is more vulnerable and prone to be hacked and the data being misused.

II. Literature Review

In 2008, when blockchain came into light it became the first decentralized cryptocurrency which not only changed the financial industry but also became an easy solution for peer-to-peer (P2P) exchange of information in the most secure, efficient, and transparent manner. As mentioned before it is a public ledger that keeps a record of all transactions in order and secure by a consensus mechanism. Some of the reviews about blockchains are mentioned in this section.

In the 2017 International Conference on Computational Intelligence and Networks, Rishav Chatterjee and Rajdeep Chatterjee have published apaper on “An Overview of the Emerging Technology: Blockchain”[1] whichspreads like onall the basic points of how exactly the blockchain works and have further spread light on the history of blockchain which is explained with the emergence of bitcoins and the man who invented bitcoin (Satoshi Nakamoto). In the paper they have also mentioned different technologies that are

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emerging with the blockchain as base such as, colored coins are the protocols which permit other digital assets to be transferred in the blockchain but not bitcoin itself by using bitcoins as tokens. Another example is the alternative blockchains wherein we can shift the bitcoins of a particular blockchain to a new blockchain. There is a new blockchain called the Ethereum blockchain which is carried out by a set of contracts called the smart contracts which is nothing but a deal between the client and the receiver. Further they have explained the working of blockchain and given a statistics on how fast the blockchain industry is growing by studying the upward trend where there are a lot of research papers on blockchain being written.

In another 2017 IEEE 6th International Congress on Big Data, Zibin Zheng, Shaoan Xie, Hongning Dai, Xiangping Chen, and Huaimin Wang published a paper on "An Overview of Blockchain Technology: Architecture, Consensus, and Future Trends"[2] as the name indicates it throws light on the blockchain architecture which is already been discussed which includes the explanation of basic working of blockchain. Further they have explained the three different types of blockchains i.e., the public blockchain where the records are visible to all, in private blockchain only certain groups who have the permission can join the consensus process and consortium blockchain where only a part of blockchain is decentralized and only those nodes are allowed to determine the consensus. And further the paper shows us the different consensus algorithms used like PBFT (Practical byzantine fault tolerance), DPOS (Delegated proof of stake), Ripple, Tendermint. And lastly the paperdepicts about the challenges faced in blockchain like scalability, privacy leakage, selfish mining.

In a 2017 IEEE European Symposium on Security and Privacy Workshops, Harry Halpin and Marta Piekarska published paper on "Introduction to Security and Privacy on the Blockchain"[3] which depicts the different areas where there should bemore security and privacy in certain areas of blockchain like the smart contracts etc., They have also discussed about different companies who have increased security through certain schemes. On a whole all the aspects of security and privacy for blockchain and cryptocurrencies are being highlighted in this paper.

III. BLOCKCHAIN TECHNOLOGY

The blockchain has many applications in travel industry but before going to that let us completely understand the core components, detailed architecture of blockchain technology along with its key characteristics and shortcomings.

A. Core components of blockchain

The blockchain setup and network operations are built upon the core components shown in Figure 2. The main components are namely asymmetric cryptography, transactions, consensus mechanism and blockchain itself which is already explained, these components are the base of a blockchain.

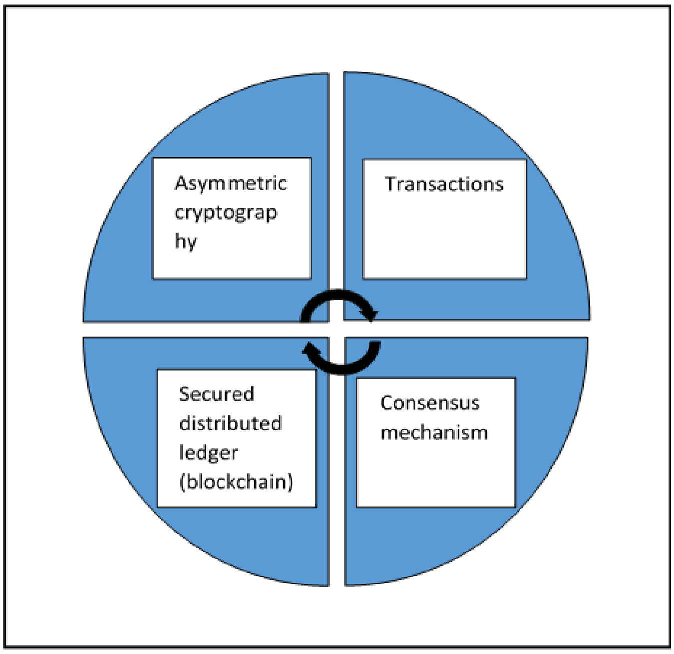

Figure 2: Core components of blockchain

1. *Asymmetric Key Cryptography*

For security, blockchain makes use of public key cryptography. For the purpose of exchanging cryptocurrencies the users need to have a digital wallet which is secured with their private key and can be only accessed with the right signatures generated using the appropriate private key. The digital wallet will have a public key known to everyone and private keys are used to sign the transactions digitally and are not visible to all except for the user himself.

2. *Transactions*

The information in blockchain is shared and exchanged among different nodes on a P2P basis. Since the entire network is interconnected, the exchange of information is done with the help of files which contains the transfer information generated by a source node which intern broadcasts to the whole network for verifying that transaction.

3. *Consensus Mechanism*

In case of blockchain it does not have a centralized system to handle the data, resolve disputes or prevent security violations for which there is a mechanism needed to regulate the flow of information, funds and also avoid frauds. All the nodes present in the chain should agree upon a common protocol for its ledger to update the appropriate content in it and the blocks cannot be simply added to the chain without the consent of majority. Hence, this is called consensus mechanism wherein the blocks are created and added to the existing ledger for the future usage.

B. Detailed architecture of Blockchain

In a blockchain, mainly to track the transactions that will occur simultaneously different transactions are grouped to form a block. It contains a block header from which each block is recognized, Hash value of previous block determines the hash value of the current block, nonce is the unique value found during proof of work, the Merkle root hash of the transactions, the timestamp, and the version of the block. To obtain the cryptographic block hash, the miners calculate the hash value of the block repeatedly which results in a random value and finally obtain a cryptographic block hash, this is called proof of work. The detailed structure of blockchain is shown in the figure 3.

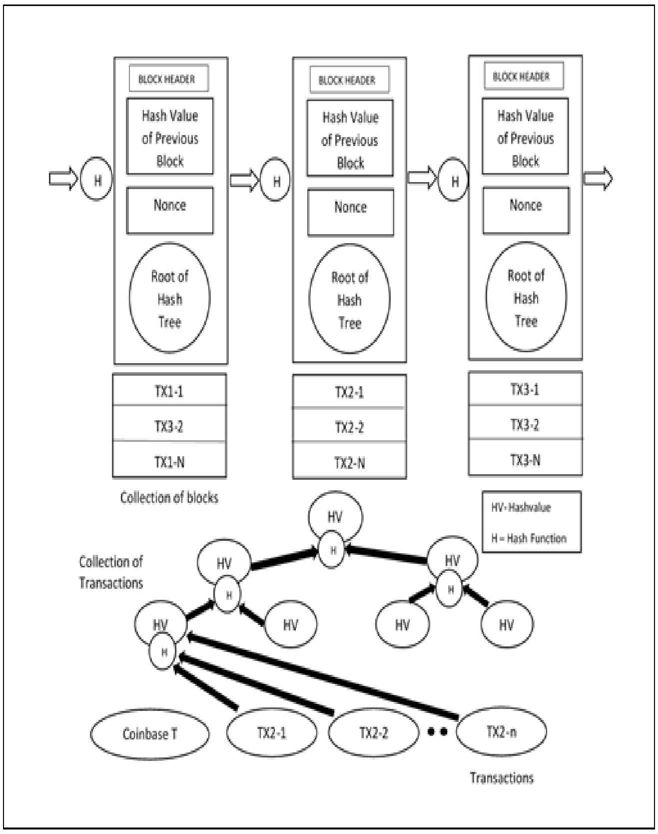

Figure 3: Detailed architecture of blockchain

As discussed before since there is no third party system, nodes should follow a consensus on whether to add or reject blocks and transactions because this should not cause problems in later stage. In case of any cryptocurrency, a consensus is achieved by proof of work which is the amount of work put in to validate the block. A cryptographic puzzle should be solved for any block to be added and shared in the ledger. This can be achieved by the nodes gathering all the verified transactions in a block and with the help of its resources like computation power and electricity, a value will be calculated that is the SHA-256 hash value of the current block.

The validation of blockchain becomes important for any transaction to be secured and fraud resistant. As we can see in figure 5, there are n number of transactions that take place in a block, each transaction gives an output which is derived by the source nodes which are authorized in the transaction of public keys. The hash generated by the public keys help to identify the correct and authentic user preserving user's privacy. In case of cryptocurrencies when they want to redeem funds they require a public key and a private key to validate them for receiving transaction outputs with the help of signatures. After all the transactions are individually validated, their chronological order will be confirmed. As discussed in figure 1, with the help of genesis block the hash value of the current block is determined by the previous block's hash value andthis is verified by the genesis block's hash value. Here for block validation, the accuracy of the timestamp is also verified along with the current block's proof of work.

C. Key characteristics of Blockchain

Blockchain has following key characteristics.

1.Decentralization. In a centralized transaction systems, all the transactions are validated in a central agency like the central bank which can cost the performance bottlenecks at thecentral servers. So contrast to this in blockchain, there is no need of third party interference. With the help of consensusalgorithm the data consistency is maintained in a distributed network within blockchain.

2.Persistency. The validation of transaction can be made very quickly and also the invalid transactions will be not admitted or added to the chain since the validation is done by some honest miners. It is impossible to revert back or delete any transaction once it is added to the blockchain. We can also discover blocks that contain invalid transactions immediately.

3.Anonymity. When interacting in blockchain with the help of a generated address, the user's real identity will be not revealed also blockchain does not guarantee perfect privacy preservation.

4.Auditability. The cryptocurrency in blockchain stores data about user balances, any transaction that is added could be easily verified and tracked.

D. Shortcomings

Blockchain has some of the limitations which are,

1. Complexity. The blockchan technology is a new addition to the market and involves completely new vocabulary. For developing there should be complex protocols applied for achieving consensus. Also it has made cryptography more mainstream, it cannot be understood overnight and implement it so easily.

2.Performance. Due to the nature of blockchain, it is basically slower than the centralised databases. The centralised databases process transactions once but in blockchain it must be processed independently by each node in a network.

3.Network Size. Blockchain basically requires large network of users. So if blockchain does not have robust network with distributed grid of nodes it becomes difficult to get full benefit of the network.

IV. Applications Of Blockchain

Blockchain has the potential to be used in any sector and one of the sectors to be discussed is the travel industry. The travel industry includes hotel reservations and airline reservations and many other but this papers focuses mainly on these two areas of application.

As the main feature of blockchain is its decentralized nature of handling data, the information can never be lost or go offline through deleting it accidently or due to any cyberattacks which ensures that the data is always traceable. In the travel industry different companies pass information/data amongst each other. For example, the travel agents who are the middlemen need to pass on the customer data to the flight companies and hotels. With the help of blockchain, storing and accessing the sensitive personal data more reliable and easier as the whole network in a blockchain will be having that information which will be broadcasted.

In the below figure 4 we can see how data in a travel sector be securely protected in a distributed network rather than to be in a potential threat.In figure 4 one potential solution is shown, where along with the travel agencies there is a distributed travel network created where all the data is being distributed among different nodes in the network. This network is not controlled by anyone which is the whole idea of a distributed network. Every participant in the node like the airline node, hotel node and the car rental node have the local copies of the entire network. The transactions can also be carried out by using thecryptocurrency in which banks and other processing centers will not be required. The data distributed will be with all the users which is immutable and cannot be tampered by any hacker.

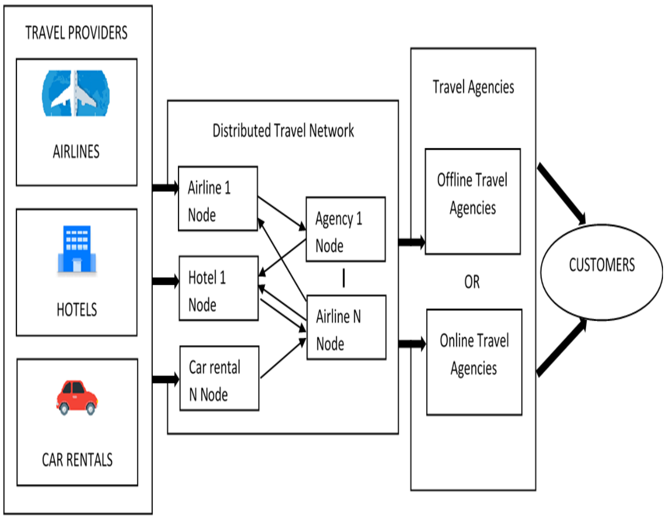

Figure 4: Blockchain in travel sector for data security

Another potential use of blockchain is the use of cryptocurrencies for easy payments in online travel bookings. During a normal online transaction for booking we encounter middle men such as the credit card company, booking sites, synchronization fee which synchronizes the booking calendar and then finally reaches the main travel provider. In this flow both the customer as well as the service provider will lose some amount of money and during this traversal of money it can be easily redirected using man in the middle attack. All this can be avoided when we use any kind of cryptocurrency where the money is transferred between A (person1) directly to B (person 2) where no one can tamper the network and no more middle men are present.

A. *Potential uses of blockchain in travel industry*

Some of the uses of blockchain technology being used in travel sector are,

1. *Identity Management*

A survey says that millions of travel documents are being recorded as lost or stolen in recent years. But, with the help of blockchain all the travel documents can be recorded and stored permanently which can be accessible easily with a single click. This eliminates the need of producing these document at several checkpoints during the customer's travel.

2. *Fraud Prevention*

During hotel bookings there can be fake websites operated by fraud people which looks like a real site for bookings which intern affects the customers. This can be avoided using blockchain where it helps verify the sites and vice versa. As the documents and customer identities will be stored in blocks the travel agencies need to check bookings that are a potential threat. Hence, all the data can be accessed securely in a blockchain and the frauds can be reduced significantly.

3. *Overbooking*

Since there is a concept of double-booking in airlines the threat of people losing their seat even after booking it is more. With the help of blockchain, the passengers have increased security for reservations since their reservations are stored in a block securely and cannot be changed. Hence, the airlines should leave the practice of double-booking.

V. Conclusion

This paper gives an overview of how blockchain works and its potential in transforming the traditional working of travel industry with the key characteristics being discussed. This paper signifies the detailed architecture of blockchain and how it works and what are the key components of blockchain. These days blockchain based applications are becoming more and more and one of it is being discussed here which is the application of blockchain in travel industry. The travel industry includes the hotel bookings and airline bookings which are the two applications discussed in this paper. Blockchain can bring huge change in the travel industry in terms of data protection, fraud prevention and quick credit transactions using any form of cryptocurrency wherein no middlemen will be present which makes the transactions easier and better.Further blockchain can be applied in various other fields such as smart contracts, public services, Internet of Things (IoT), reputation systems and security services.

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